

SOURAV DATTO

Ph.D. Applicant - Computer Science | AI/ML, Quantum Machine Learning, Network Optimization

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RESEARCH PROFILE

Computer Science and Engineering graduate from American International University-Bangladesh (CGPA: 3.65/4.00) with research experience in machine learning, quantum machine learning, reinforcement learning, network/security analytics, and optimization-oriented AI systems. Seeking to pursue doctoral research on AI-driven quantum networking, satellite-assisted entanglement distribution, fidelity-aware resource allocation, and optimization for future Quantum Internet infrastructures.

EDUCATION

American International University-Bangladesh (AIUB)

B.Sc. in Computer Science and Engineering, Major: Information Systems

Relevant coursework: Calculus I-II, Discrete Mathematics, Data Structures, Algorithms, Computer Networks, Machine Learning, Data Science, Computer Vision and Pattern Recognition, Linear Programming, Network Security, Wireless Sensor Networks.

Completed February 2026

CGPA: 3.65/4.00

148 credits, 56 courses

RESEARCH INTERESTS

Quantum networking and communications; satellite-assisted quantum networks; AI for quantum systems; entanglement routing and scheduling; fidelity-aware resource allocation; network optimization; machine learning for distributed systems; reinforcement learning for control and scheduling.

RESEARCH EXPERIENCE

Research Assistant | Research Centre for Green Energy, Electro-Anatomy, and Digital Intelligence - Bangladesh Oct 2024 - Present

- Conduct applied AI research on smart energy networks, stability prediction, and healthcare intelligence systems.
- Develop hybrid machine learning frameworks for predictive analytics, classification, and stability-oriented decision support.
- Built research workflows involving statistical modeling, data preprocessing, reproducible experiments, and technical writing.

Lead Researcher | Tech Wing Lab - Bangladesh Jun 2024 - Present

- Lead small research teams working across ML, NLP, computer vision, cybersecurity, and quantum machine learning projects.
- Supervise junior researchers on dataset preparation, model evaluation, paper drafting, visualization, and conference submission workflows.
- Prepared and presented work for international conference venues including ICCIT, ICERIE, ICCIES, ICDSAIA, and TENCON.

SELECTED PUBLICATIONS AND RESEARCH MANUSCRIPTS

- Sourav Datto, Kazi Redwan, Arifa Sultana, Delower Hossen Tuhin, Sadik Saleh, and Md. Faruk Abdullah Al Sohan. "Quantum Kernel Learning for Multiclass Pattern Recognition in Structured Data." Accepted for presentation, IEEE TENCON 2025; first author; led experiments and implementation. Status note: registration/presentation not completed due to funding constraints.
- "Dynamic and Transformative Hybrid Machine Learning Strategies for Effective Stability Prediction in Smart Energy Networks." ICCIES/ICDSAIA 2024, IEEE Xplore.
- "Integrating Machine Learning and Clinical Expertise: A Comparative Study for Improved Breast Cancer Diagnosis." ICCIES/ICDSAIA 2024, IEEE Xplore.
- "Melanoma Detection Using Augmented ResNet34 for High-Precision Dermoscopic Image Classification." ICCIES/ICDSAIA 2025, Springer.
- "Reinforcement Learning for Smart Grid Stability Using Adaptive Control and State Abstraction." In press/manuscript.
- "Enhanced Phishing Payload Detection Using Fine-Tuned DistilBERT and XAI-Based NLP Models." In press/manuscript.
- "A Comparative Study of Machine Learning Models for Cardiovascular Risk Prediction Using PCA-Transformed Framingham Data." In press/manuscript.

SELECTED RESEARCH PROJECTS

- **Quantum Kernel Learning (QSVC)** - Implemented QSVC with PauliFeatureMap, quantum kernel fidelity analysis, Kernel PCA visualization, circuit-depth evaluation, and noise simulation; achieved 93.33% accuracy on multiclass structured data.
- **Reinforcement Learning for Smart Grid Stability** - Designed Q-learning/state-abstraction approach for adaptive control and stability prediction; relevant to scheduling/resource-allocation problems under system uncertainty.
- **SecOps Log Anomaly Triage for Distributed Systems** - Built an unsupervised HDFS log triage pipeline using Isolation Forest, session-level feature engineering, rare-event/transition analysis, hybrid risk scoring, severity labels, and analyst explanations.

- **Transformer-Based Phishing Email Detection** - Fine-tuned DistilBERT and benchmarked CLS embeddings with classical ML; reported 98.52% accuracy and 0.9993 ROC-AUC in the original notebook evaluation.
- **LLM Prompt Injection Detection Benchmark** - Constructed a comparative AI-security benchmark using official train/test splits, BoW/TF-IDF/embedding features, hyperparameter tuning, and final test evaluation.

TECHNICAL SKILLS

Programming & Data Science	Python, C++, R, SQL, NumPy, Pandas, scikit-learn, statistical modeling, data analysis
Machine Learning & AI	Supervised/unsupervised ML, deep learning, NLP, computer vision, reinforcement learning, anomaly detection, explainable AI
Quantum & Optimization	Quantum machine learning, QSVC, quantum kernels, Qiskit, PauliFeatureMap, kernel methods, network/resource optimization concepts
Tools	PyTorch, TensorFlow/Keras, Hugging Face Transformers, Matplotlib, Plotly, Git/GitHub, Jupyter, LaTeX/IEEE templates

CONFERENCE PRESENTATIONS

- 27th International Conference on Computer and Information Technology (ICCIT 2024) - Cox's Bazar, Bangladesh
- 8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025) - SUST, Sylhet, Bangladesh
- International Conference on Data Science, AI, and Applications (ICDSAIA 2025) - Dhaka, Bangladesh
- 1st International Conference on Computational Intelligence in Engineering Science (2025) - Ho Chi Minh City, Vietnam
- IEEE TENCON 2025 - accepted paper notification for Quantum Kernel Learning; final participation constrained by funding

SELECTED ACHIEVEMENTS

- Data Innovators Challenge (MetaData 2024) - Finalist
- Project Showcasing - Robo Carnival 2024, BUET Robotics Society
- IEEE Student Professional Awareness Venture (SPAve) 7.0 - 2024
- Game of Datathon (Bitfest 2025) - Participant

REFERENCES

Md. Faruk Abdullah Al Sohan, Lecturer, Department of Computer Science, Faculty of Science & Technology, AIUB - faruk.sohan@aiub.edu
Abu Shufian, Lecturer, Department of Electrical & Electronic Engineering, AIUB - shufian.eee@aiub.edu
Third academic/professional recommender: to be confirmed before KSU application submission.